LAST NAME: _____

Practice Quiz 10D — MATH 2055 — Fall 2026

Practice for Quiz 10

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Let $a_1 = 1$, $a_2 = 2$, and

$$a_n = a_{n-1} + 2a_{n-2} \qquad (n \geq 3).$$

A student claims $a_n = 2^{n-1}$ for every integer $n \geq 1$ and writes:

> The formula holds at $n = 1$. Assume $a_n = 2^{n-1}$. Then
>
> $$a_{n+1} = a_n + 2a_{n-1} = 2^{n-1} + 2 \cdot 2^{n-2} = 2^n.$$
>
> Thus induction proves the formula.

(a) Identify the missing base case and the unsupported use of the inductive hypothesis.

(b) Give a corrected proof using strong induction.